Difference Equation System Descriptions

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Key Concepts

1) A difference equation is the discrete-time version of a differential equation.

2) The present output of a difference equation is a weighted sum of past values of the output and present/past values of the input.

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

or

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \quad (a_0 = 1)$$

a) N is the order of the system.

b) The parameters $\{a_k, b_k\}$ define the system characteristics.

3) Difference equations are widely used in computational applications to implement signal-processing algorithms on data.

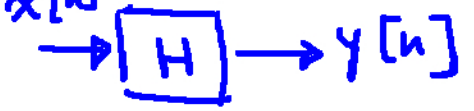
4) Given the input and the initial values of the output (often assumed zero), the rest of the output values are found recursively.

5) Different choices for the coefficients in the difference equation produce different effects, such as smoothing the signal or eliminating slow variations.

Video Screenshots

Discrete time analog of differential equation

"King of computation" for system outputs



$$y[n] + a_1 y[n-1] + a_2 y[n-2] + \dots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M]$$

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \qquad (a_0 = 1)$$

N^{th} order

$N+M+1$ parameters $\{a_k, b_k\}$ completely define the system

Computation

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

present output

past outputs

input

Example: $y[n] - \frac{1}{2}y[n-1] = \frac{1}{4}x[n] + \frac{1}{4}x[n-1]$

Rewrite: $y[n] = \frac{1}{2}y[n-1] + \frac{1}{4}x[n] + \frac{1}{4}x[n-1]$

Assume start with $n = 0$

$y[0] = \frac{1}{2}y[-1] + \frac{1}{4}x[0] + \frac{1}{4}x[-1]$

↑ initial condition

Iterate

$y[1] = \frac{1}{2}y[0] + \frac{1}{4}x[1] + \frac{1}{4}x[0]$

$y[2] = \frac{1}{2}y[1] + \frac{1}{4}x[2] + \frac{1}{4}x[1]$

⋮

MATLAB "filter"

Suppose $x[n] = \begin{cases} 0 & n<0 \\ 1 & n \geq 0 \end{cases}$

$y[-1] = 0$

$y[0] = \frac{1}{2}\cdot 0 + \frac{1}{4}\cdot 1 + \frac{1}{4}\cdot 0 = \frac{1}{4}$

$y[1] = \frac{1}{2}\cdot\frac{1}{4} + \frac{1}{4}\cdot 1 + \frac{1}{4}\cdot 1 = \frac{5}{8}$

$y[2] = \frac{1}{2}\cdot\frac{5}{8} + \frac{1}{4}\cdot 1 + \frac{1}{4}\cdot 1 = \frac{13}{16}$

$y[3] = \frac{1}{2}\cdot\frac{13}{16} + \frac{1}{4}\cdot 1 + \frac{1}{4}\cdot 1 = \frac{29}{32}$

⋮

Four Examples:

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1) 6 point averaging

$$y[n] = \frac{1}{6}\{x[n] + x[n-1] + \cdots + x[n-5]\}$$

2) 6 point differencing

$$y[n] = \frac{1}{6}\{x[n] - x[n-1] + x[n-2] - \cdots - x[n-5]\}$$

3) Recursive LowPass

$$y[n] = 0.95\,y[n-1] + 0.05\,x[n]$$

4) Recursive High Pass

$$y[n] = -0.95\,y[n-1] + 0.05\,x[n]$$

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Inputs:

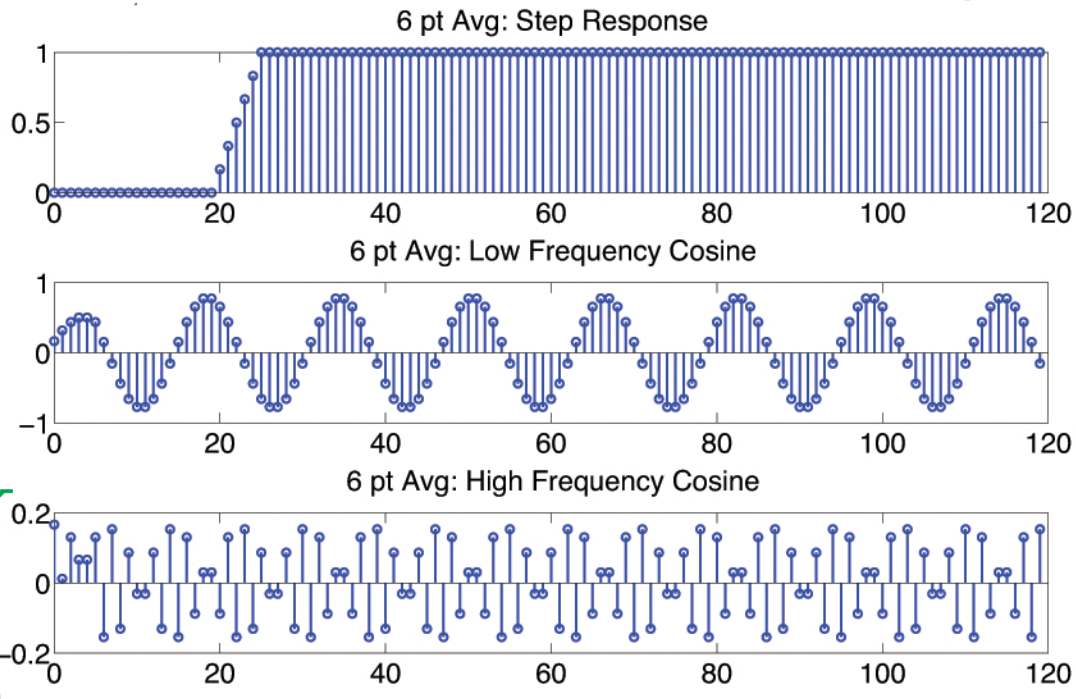
a) $x[n] = \begin{cases} 0 & n < 20 \\ 1 & n \geq 20 \end{cases}$

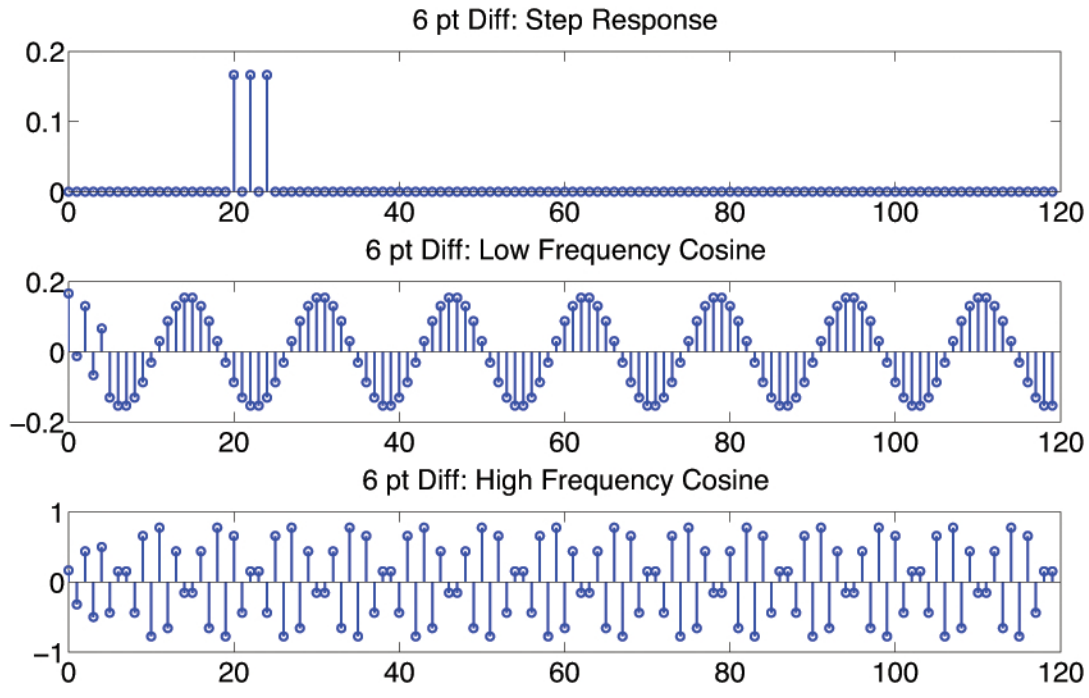
b) $x[n] = \begin{cases} 0 & n < 0 \\ \cos(\pi n/8) & n \geq 0 \end{cases}$

c) $x[n] = \begin{cases} 0 & n < 0 \\ \cos(7\pi n/8) & n \geq 0 \end{cases}$

$y[n] = \frac{1}{6}\{x[n] + x[n-1] + \dots x[n-5]\}$

$y[n] = \frac{1}{6}\{x[n] - x[n-1] + x[n-2] - \dots - x[n-5]\}$





Inputs:

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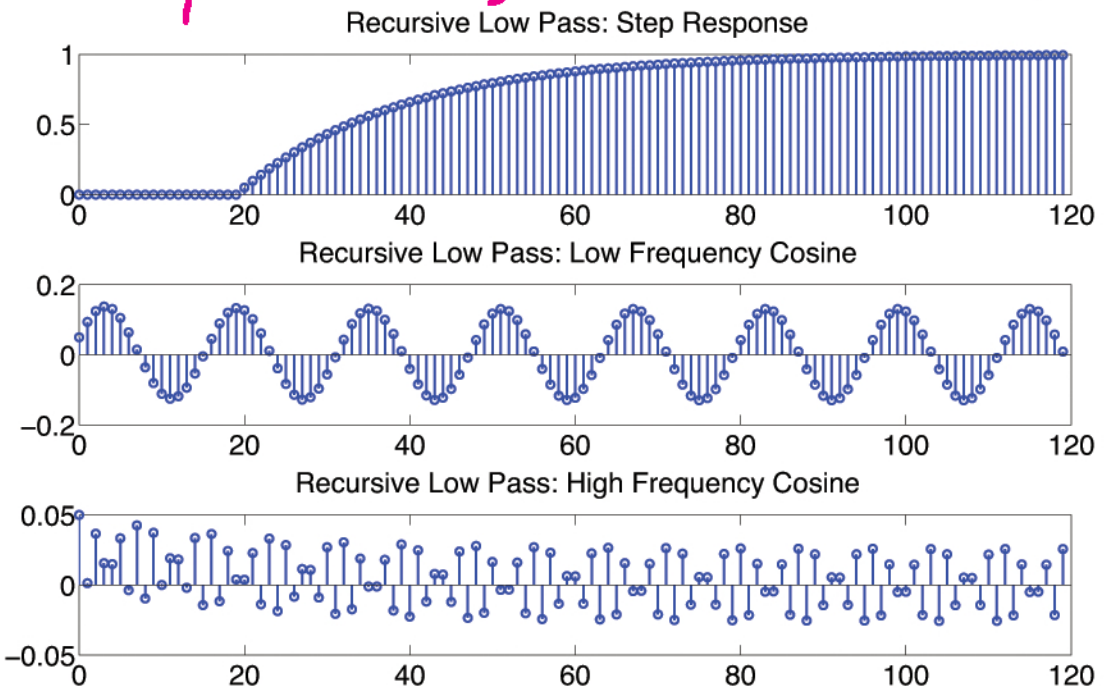
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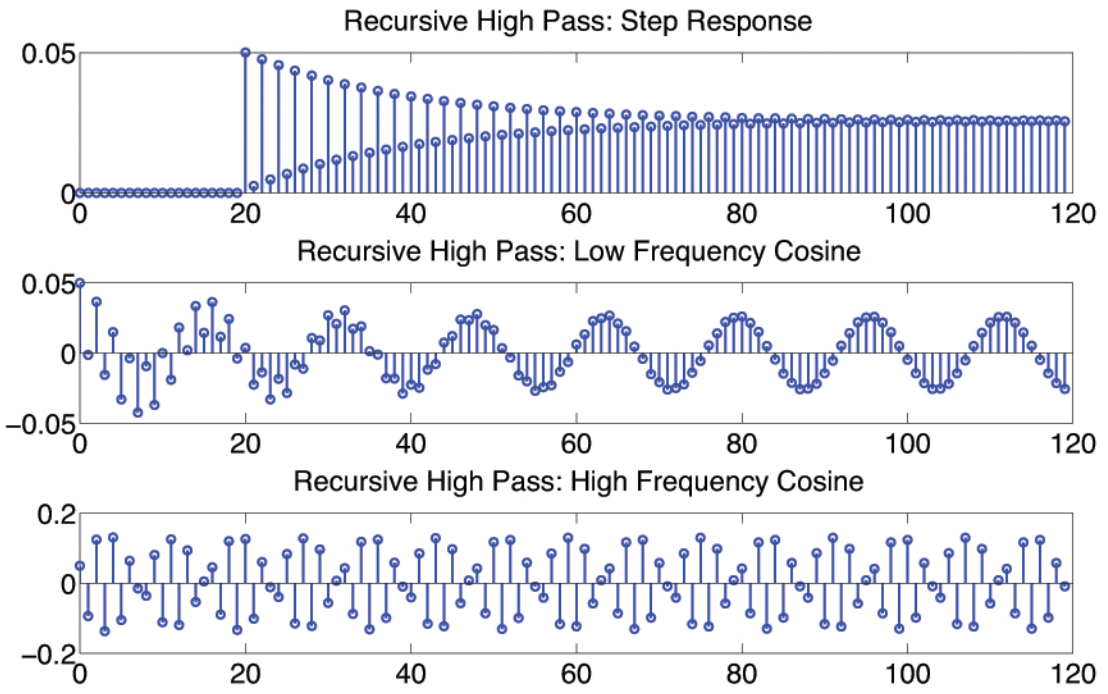
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$y[n] = 0.95\,y[n-1] + 0.05\,x[n]$



$y[n] = -0.95\,y[n-1] + 0.05\,x[n]$



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